Cinematography: Camera Placement/Movement

Special Effects

Special effects is a general term reserved for technology used to create images that would be too dangerous, too expensive, or simply impossible to achieve with traditional cinematographic approaches. For audiences, a major attraction of movies has always been their ability to create illusion.

As is often the case with movie terminology, the names used to categorize special effects are somewhat convoluted. In this case, we can blame the vagaries on the evolution from film to digital capabilities. During the celluloid era, mechanical effects were those created and photographed on set; optical effects were created by manipulating the image and/or film negative "in-camera" during production and/or during the film-stock processing after the negative had been exposed.

Mechanical Effects

These days, when the postproduction process (even for those few films shot on film stock) is completed on computers, the term visual effects refers to those effects created and integrated using computers in postproduction. The current specific use of *special effects* is synonymous with mechanical effects: any effect generated on set that can be photographed by the camera. These effects may also be referred to as *practical effects*.

For their farcical but strangely moving buddy movie *Swiss Army Man* (2016), directors Daniel Kwan and Daniel Scheinert (known collectively as "the Daniels") innovated a series of relatively low-tech mechanical effects that made Manny, the soulful and supernaturally resourceful corpse played by Daniel Radcliffe, to appear to spray gallons of water and shoot rocks from his mouth, chop logs with his arm, and set himself on fire. Multiple life-size dummies were made from molds of Radcliffe for different purposes: dropping and throwing, carrying, and—for the film's infamous first scene—using simulated flatulence for the corpse to appear to propel itself through the water.

***Swiss Army Man* (Daniel Kwan and Daniel Scheinert, 2016)**

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Optical Effects

Optical effects in the celluloid era could be as simple as the “stop trick” used to depict the aforementioned queen’s execution or as complicated as the Schüfftan process (named for its inventor, Eugen Schüfftan), which integrated actors onto shots of sets built in miniature by using etched mirrors to expose the two very different elements precisely onto the same image. Other optical effects include the similar but simpler matte shot, which exposed an image onto one portion of the negative but kept another portion unexposed so it could be filled with an exposure of another element later. *The Great Train Robbery* (1903) featured an early example of this technique to show the passing countryside through the window of a baggage car that was stationary when the scene was filmed.

For Fritz Lang’s *Metropolis* (1927), a pioneering science-fiction film, the city of the future was a model created by designer Otto Hunte. Special effects photography turned this miniature into a massive place on-screen, filled with awe-inspiring objects and vistas.

***Metropolis* (Fritz Lang, 1927)**

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(CGI) Computer Generated Imagery

In the digital age, computer-generated imagery (CGI) has largely eclipsed optical effects, replacing them with visual effects that can create settings and backgrounds with more accuracy and less cost. CGI backgrounds are often used for spectacular imaginary worlds in fantasy and science-fiction films, but are also used to enhance backgrounds to allow scenes set in contemporary (and even relatively mundane) locales to be shot on a soundstage or at a more convenient location. Actors can be placed within digitally generated settings by filming the actors against a uniformly colored backdrop (usually bright green, hence the term *green screen*) and then applying chroma keying, a process that digitally removes that color so it can be replaced with computer-generated images.

Director Denis Villeneuve had a huge budget and state-of-the-art visual effects at his disposal for the sequel *Blade Runner 2049* (2017), but he still avoided CGI and the green screen whenever possible because it denies actors the opportunity to interact organically with their surroundings. His point is ironically illustrated in the shot above, in which an actor on a physical set faces a superimposed video projection. Both *Blade Runner* films examine the distinctions between what is artificial and what is real, a theme visualized in the way color, relative size, and a sort of variegated video-texture is applied to the simulated woman addressing the protagonist—who is himself a simulated human.

***Blade Runner 2049* (Denis Villeneuve, 2017)**

Motion Capture

Motion capture (also known as motion tracking or mocap) is a specific visual effect in which a live-action subject wears a bodysuit fitted with reflective markers that enables a computer to record each movement as digital images; they are then translated, with as much manipulation as desired, into models on which the screen figures are based. When the images include facial contours and expressions, the process is called *performance capture*. As spectacular as some special effects can be, however, the goal of virtually all of the effects previously described is almost always to create verisimilitude—an illusion of reality or a believable alternative reality.

Benedict Cumberbatch didn't simply provide the voice for Smaug, the colossal psychotic dragon in *The Hobbit: The Desolation of Smaug* (2013). Director Peter Jackson and his second unit director, Andy Serkis (the actor who created Gollum, considered cinema's first truly convincing cybercharacter), enlisted Cumberbatch to use motion-capture and performance-capture technology to inform the 3-D computer animators as to the dragon's movements and expressions.

***The Hobbit: The Desolation of Smaug* (Peter Jackson, 2013)**